

Linear Models Lecture 3: Estimation of the Linear Projection Model

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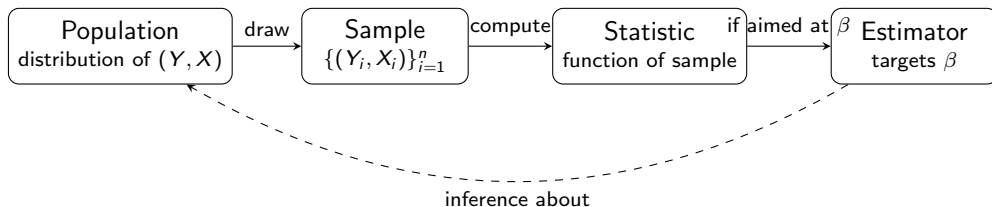
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Deriving the OLS estimator

- Recall, geometry can work in populations or samples.
- In the following slides we will derive
 - the OLS slope estimator $\hat{\beta}$,
 - the OLS estimator for error variance,
 - Decomposition of OLS (Analysis of Variance), with R^2
- Next time, the FWL Theorem.

Review: From Population to Inference

- The CEF and linear projection are facts about **populations**—we derived $\beta = [\mathbb{E}[XX']]^{-1}\mathbb{E}[XY]$ last time.
- But we don't observe populations. We observe **samples**.



- A **statistic** is any function of the sample (e.g. $\bar{Y}$, s^2 , a histogram).
- An **estimator** is a statistic designed to estimate a population parameter.

Samples

- We estimate the projection coefficient β with measurements of (Y, X) , a *sample*.
- $\{(Y_i, X_i) : i \in 1, \dots, n\}$, is the sample, governed by the distribution of (Y, X) .
- From the next slide forward, $\mathbf{X}$ is going to be a matrix, and we will use lower case for vectors e.g. $\mathbf{e}$.

Identically Distributed.

- Main approach is to assume homogeneity: $\{(Y_i, \mathbf{x}_i) : i \in 1, \dots, n\}$ are identically distributed.

$$\begin{aligned} \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} &= \beta_1 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \beta_2 \begin{bmatrix} X_{21} \\ \vdots \\ X_{2n} \end{bmatrix} + \beta_3 \begin{bmatrix} X_{31} \\ \vdots \\ X_{3n} \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} \\ &= \begin{bmatrix} 1 & X_{21} & X_{31} \\ \vdots & \vdots & \vdots \\ 1 & X_{2n} & X_{3n} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} \\ \mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \mathbf{e} \end{aligned}$$

- Note, $\mathbb{E}[Y_1|\mathbf{x}_1] \neq \mathbb{E}[Y_2|\mathbf{x}_2]$ in general (different covariates imply different conditional means).

Least Squares Estimator

- The linear projection coefficient β is the minimizer of the expected squared error

$$S(\beta) = \mathbb{E}[(Y - \mathbf{x}'\beta)^2]$$

- The *moment estimator* of $S(\beta)$ is the sample average:

$$\hat{S}(\beta) = \frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{x}_i'\beta)^2$$

- $\sum_{i=1}^n (Y_i - \mathbf{x}_i'\beta)^2$ is called $\text{SSE}(\beta)$, the **sum of squared errors**.
- The estimator $\hat{\beta}$ is the minimizer of $\hat{S}(\beta)$, as well as the minimizer of the SSE.
- $\hat{\beta}$ is called the least squares estimator, sometimes written $\hat{\beta}_n$.

The Plan

- We have a population quantity β and a sample objective $\hat{S}(\beta)$. Now we need to:
 - 1 **Find** $\hat{\beta}$: minimize SSE using matrix calculus.
 - 2 **Understand** $\hat{\beta}$ geometrically: projection matrices $\mathbf{P}$ and $\mathbf{M}$.
 - 3 **Quantify uncertainty**: estimate σ^2 from residuals, then get standard errors for $\hat{\beta}$.
- Note, residuals $\neq$ disturbances. The former is estimated.

Finite Sample Assumptions of OLS estimator

- Assumption 1: The random variables $\{(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)\}$ are independent and identically distributed.
- Assumption 2: $Y = \mathbf{X}'\boldsymbol{\beta} + e$, where $\mathbb{E}(e|\mathbf{X}) = 0$.
- Assumption 3: $\mathbb{E}(\mathbf{X}\mathbf{X}') > 0$ (with probability 1).
- Assumption 4*: $\mathbb{E}[e^2|\mathbf{X}] = \sigma^2(\mathbf{X}) = \sigma^2$.

Derivation of Least Squares Estimator

- The functional form of the least squares estimator can be derived via multivariate calculus.
- We look for a minimum of $SSE(\beta)$.
- Take first derivatives to calculate first-order conditions and locate the critical values.
- Then we check we have the global minimum, by using the fact that positive definite matrices represent convex functions.

Calculus Derivation of Least Squares Estimator

$$\begin{aligned}SSE(\beta) &= (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) && \text{(Definition of } \mathbf{e} \text{)} \\&= (\mathbf{y}' - \beta' \mathbf{X}')(\mathbf{y} - \mathbf{X}\beta) && \text{(Transpose rules)} \\&= \mathbf{y}'\mathbf{y} - \beta' \mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta + \beta' \mathbf{X}'\mathbf{X}\beta && \text{(Distributive property)} \\ \min_{\beta} SSE(\beta) &= \min_{\beta} \underbrace{\mathbf{y}'\mathbf{y}}_{\text{constant}} - \underbrace{2\mathbf{y}'\mathbf{X}\beta}_{\text{linear in } \beta} + \underbrace{\beta' \mathbf{X}'\mathbf{X}\beta}_{\text{quadratic form}}\end{aligned}$$

Calculus Derivation of Least Squares Estimator

$$\begin{aligned}\frac{\partial \mathbf{y}'\mathbf{y} - \beta'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta + \beta'\mathbf{X}'\mathbf{X}\beta}{\partial \beta} &= \mathbf{0} - \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{y} + (\mathbf{X}'\mathbf{X} + \mathbf{X}'\mathbf{X})\hat{\beta} \\ &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} && \text{(Symmetric)} \\ -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} &= \mathbf{0} && \text{(find critical point)} \\ \mathbf{X}'\mathbf{X}\hat{\beta} &= \mathbf{X}'\mathbf{y}. && \text{(normal equations)} \\ \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{(If } \mathbf{X}'\mathbf{X} \text{ can be inverted (A3).)}$$

Second order condition

Take second derivative, and check that it is positive

$$\frac{\partial^2 -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta}}{\partial \hat{\beta}'} = 2(\mathbf{X}'\mathbf{X})'$$

Which is positive so long as $\mathbf{X}'\mathbf{X}$ is positive definite (Assumption A3).
 $\hat{\beta}$ minimizes the sum of squared errors, assuming $\mathbf{X}$ is of full rank.

OLS in R: The Formula in Action

```
library(carData)
X <- cbind(1, Prestige$education, Prestige$income)
y <- Prestige$prestige

# The matrix formula from the slides:
solve(t(X) %*% X) %*% t(X) %*% y

# What lm() computes:
coef(lm(prestige ~ education + income, data=Prestige))
```

The two are identical: $\text{lm}()$ implements $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

Projection Matrix

- There is a geometric interpretation for the minimization problem that uses the idea of “projection”.
- The following matrix is called a “projection matrix”:

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

- Intuition: $\mathbf{P}$ takes any vector in $\mathbb{R}^n$ and drops it onto the column space of $\mathbf{X}$ —the closest point that can be written as $\mathbf{X}b$ for some b .
- The key payoff: $\mathbf{P}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\mathbf{y}}$. This is why $\mathbf{P}$ is also called the “hat matrix”—it puts the hat on $\mathbf{y}$.

Properties of the Projection Matrix

- P leaves anything already in the column space alone:
 - $PX = X(X'X)^{-1}X'X = X$
 - If $X = [X_1 \ X_2]$, then $PX_1 = X_1$.
- P is **symmetric**: $P = P'$.
- P is **idempotent**: $PP = P$.
- These two properties (symmetric + idempotent) will do a lot of work for us in the rest of the course.

Projection onto column of 1

- If $\mathbf{X} = \mathbf{1}$, then $\mathbf{P}_1 = \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}' = \frac{1}{n}\mathbf{1}\mathbf{1}'$
- So if we project $\mathbf{y}$ onto $\mathbf{P}_1$, we get a vector repeating the sample mean:

$$\mathbf{P}_1\mathbf{y} = \frac{1}{n}\mathbf{1}\mathbf{1}'\mathbf{y} = \mathbf{1} \cdot \frac{1}{n} \sum_{i=1}^n Y_i = \mathbf{1}\bar{Y}$$

- **Takeaway:** The simplest regression (intercept only) projects $\mathbf{y}$ onto the sample mean. Every additional regressor refines this projection.
- The annihilator $\mathbf{M}_1\mathbf{y} = \mathbf{y} - \mathbf{1}\bar{Y}$ *demeans* $\mathbf{y}$ —it removes the part explained by the intercept.

Orthogonal Projection

- The following matrix is called an "orthogonal projection matrix", or the Annihilator Matrix.

$$\begin{aligned} \mathbf{M} &= \mathbf{I}_n - \mathbf{P} \\ &= \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \end{aligned}$$

- $\mathbf{M}$ and $\mathbf{X}$ are orthogonal:

$$\begin{aligned} \mathbf{MX} &= (\mathbf{I}_n - \mathbf{P})\mathbf{X} \\ &= \mathbf{X} - \mathbf{PX} \\ &= \mathbf{X} - \mathbf{X} \\ &= \mathbf{0} \end{aligned}$$

- We can define the residuals from projecting onto a column of 1s: $\mathbf{M}_1\mathbf{y} = \mathbf{y} - \mathbf{1}\bar{Y}$, this demeans $\mathbf{y}$.

Relationship between residuals and disturbances

$$\begin{aligned}\hat{e} &= \mathbf{y} - \hat{\mathbf{y}} \\ &= \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} \\ &= \mathbf{y} - \mathbf{P}\mathbf{y} \\ &= \mathbf{M}\mathbf{y} \\ &= \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \mathbf{e}) \\ &= \mathbf{M}\mathbf{X}\boldsymbol{\beta} + \mathbf{M}\mathbf{e} \\ &= \mathbf{M}\mathbf{e}\end{aligned}$$

Projection and Residuals in R

```
mod <- lm(prestige ~ education + income, data=Prestige)
X <- cbind(1, Prestige$education, Prestige$income)
P <- X %*% solve(t(X) %*% X) %*% t(X)
M <- diag(nrow(X)) - P

# P*y = fitted values
all.equal(as.vector(P %*% y), fitted(mod)) # TRUE

# M*y = residuals
all.equal(as.vector(M %*% y), resid(mod)) # TRUE
```

P and M are used by R to compute `fitted()` and `resid()`.

Where We Stand

- We have $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and the geometry behind it ($\mathbf{P}$, $\mathbf{M}$).
- But how precise is $\hat{\beta}$? To answer that, we need $\text{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.
- Problem: σ^2 is *unknown*—it's a population quantity, just like β .
- So we need a second estimator, s_e^2 , built from the residuals. This is step 3.

Why Does Projection Help?

- We want to estimate the systematic part $\mathbf{X}\beta$. Two candidates:
 - 1 Use $\mathbf{y}$ itself (just report the raw data).
 - 2 Use $\mathbf{Py} = \hat{\mathbf{y}}$ (the projection onto the column space of $\mathbf{X}$).
- Compare their variances as estimators of $\mathbf{X}\beta$:

$$\begin{aligned}\text{Var}(\mathbf{y}|\mathbf{X}) &= \sigma^2 \mathbf{I} \\ \text{Var}(\mathbf{Py}|\mathbf{X}) &= \sigma^2 \mathbf{P}\end{aligned}$$

- The difference is $\sigma^2(\mathbf{I} - \mathbf{P}) = \sigma^2 \mathbf{M}$, which is positive semi-definite.
- So $\text{Var}(\mathbf{y}) \geq \text{Var}(\mathbf{Py})$: projection *removes noise* without distorting the signal, because $\mathbf{PX}\beta = \mathbf{X}\beta$.

Variance of the regression errors

- We want to measure the precision of our regression estimates.
- The error (or disturbance) of an observation is the deviation of a value from its theoretical mean, cannot be observed.
- The coefficients inherit that uncertainty from the theoretical model.
- Our estimates have additional uncertainty, related to the fact we only have a sample.
- Residuals are the difference between observations and estimates, can be observed.

The Bane of Statistics: We Never See the Errors

	Disturbance e_i	Residual $\hat{e}_i$
Definition	$Y_i - \mathbf{x}_i' \beta$	$Y_i - \mathbf{x}_i' \hat{\beta}$
Depends on	population β	estimated $\hat{\beta}$
Observable?	No	Yes
Distribution	$E[e_i \mathbf{X}] = 0, \text{Var}(e_i) = \sigma^2$	$\hat{e}_i = \mathbf{M}_i \mathbf{e}$ (a linear combo of <i>all</i> e_j 's)

- We showed that $\hat{\mathbf{e}} = \mathbf{M}\mathbf{e}$: each residual is a *filtered* version of the true disturbances, not a direct copy.
- This means MSE (computed on residuals) $\neq$ true error variance. We will need a degrees-of-freedom correction ($n - k$) to fix this.

Estimating the Disturbance Variance: First Attempt

- Terminology note: *disturbance*, *error*, and *noise* all refer to $e_i = Y_i - \mathbf{x}_i'\beta$. We use them interchangeably; the key point is that e_i is **unobservable**.
- Natural idea: plug the residuals $\hat{e}_i$ into the sample variance formula:

$$\hat{\sigma}^2 = n^{-1}\hat{\mathbf{e}}'\hat{\mathbf{e}} = n^{-1}(\mathbf{M}\mathbf{y})'(\mathbf{M}\mathbf{y}) = n^{-1}\mathbf{y}'\mathbf{M}\mathbf{y} = n^{-1}\mathbf{e}'\mathbf{M}\mathbf{e}$$

where the last step uses $\mathbf{M} = \mathbf{M}'\mathbf{M}$ and $\mathbf{M}\mathbf{X} = \mathbf{0}$.

- But $\hat{\mathbf{e}}'\hat{\mathbf{e}} \leq \mathbf{e}'\mathbf{e}$, so residuals underestimate the true sum of squared errors:

$$n^{-1}\mathbf{e}'\mathbf{e} - n^{-1}\mathbf{e}'\mathbf{M}\mathbf{e} = n^{-1}\mathbf{e}'\mathbf{P}\mathbf{e} \geq 0$$

because $\mathbf{P}$ is positive semi-definite.

- Projection “absorbs” some of the disturbance into fitted values, so residuals understate the true noise.

The Degrees-of-Freedom Correction

- Fix: divide by $n - k$ instead of n . Define $s_e^2 = \hat{\mathbf{e}}' \hat{\mathbf{e}} / (n - k)$.
- Proof that s_e^2 is unbiased for σ^2 :

$$\begin{aligned}\hat{\mathbf{e}}' \hat{\mathbf{e}} &= \mathbf{e}' \mathbf{M}' \mathbf{M} \mathbf{e} = \mathbf{e}' \mathbf{M} \mathbf{e} \\ E[\hat{\mathbf{e}}' \hat{\mathbf{e}} | \mathbf{X}] &= E[\mathbf{e}' \mathbf{M} \mathbf{e} | \mathbf{X}] \\ &= E[\text{tr}(\mathbf{e}' \mathbf{M} \mathbf{e}) | \mathbf{X}] \\ &= E[\text{tr}(\mathbf{M} \mathbf{e} \mathbf{e}') | \mathbf{X}] \\ &= \text{tr}(\mathbf{M} E[\mathbf{e} \mathbf{e}' | \mathbf{X}]) \\ &= \text{tr}(\mathbf{M} \sigma^2 \mathbf{I}) = \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n - k)\end{aligned}$$

($\mathbf{M}$ symmetric, idempotent)

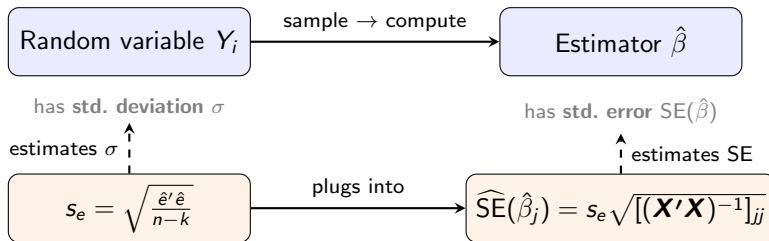
(scalar = its own trace)

($\text{tr}(\mathbf{A} \mathbf{B} \mathbf{C}) = \text{tr}(\mathbf{C} \mathbf{A} \mathbf{B})$)

($\mathbf{M}$ fixed given $\mathbf{X}$)

(A4, $\text{tr}(\mathbf{M}) = n - k$)

Standard Deviation vs. Standard Error



- **Standard deviation:** spread of a random variable (σ).
- **Standard error:** spread of an *estimator* across repeated samples.
- The punchline: to estimate how uncertain $\hat{\beta}$ is, we first need to estimate how noisy Y is. Estimation all the way down.

Interpretation of Residual Standard Error

- `summary(lm())` reports s_e as the Residual standard error or "sigma"
- STATA reports this quantity as "Root Mean Squared Error".
- s_e is on the same scale as the outcome.
- If $y > 0$, it can make sense to calculate an *average prediction error rate*: $\frac{s_e}{\bar{y}}$.
- Example: if we are predicting houses that are on average 1 million dollars, being generally off by \$50k is less bad than if we are predicting houses that are \$100k.

What Are Standard Errors For?

Consider a regression of occupational prestige on education and income:

	Estimate	Std. Error	t value	$\Pr(> t)$
(Intercept)	-6.06	4.27	-1.42	0.159
education	4.19	0.39	10.78	< 0.001
income	0.0013	0.0003	5.19	< 0.001
Residual std. error: $s_e = 7.81$ on 99 d.f. $R^2 = 0.80$				

- The **Std. Error** column tells us how much $\hat{\beta}_j$ would vary across repeated samples.
- Education: estimate = 4.19, SE = 0.39. The estimate is about 10 standard errors from zero—hard to explain by sampling noise alone.
- Where does each SE come from? From s_e and $(\mathbf{X}'\mathbf{X})^{-1}$. Let's derive that.

Variance of $\hat{\beta}$

$$\begin{aligned}
 V(\hat{\beta}|\mathbf{X}) &= V((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}|\mathbf{X}) && \text{(Normal Equation)} \\
 &= V(\mathbf{A}\mathbf{y}|\mathbf{X}) && \text{(Pick } \mathbf{A} \text{ as placeholder)} \\
 &= \mathbf{A}V(\mathbf{y}|\mathbf{X})\mathbf{A}' && \text{(Var}(cX) = c^2\text{Var}(X)) \\
 &= \mathbf{A}\Sigma\mathbf{A}' \\
 &= \mathbf{A}(\sigma^2\mathbf{I})\mathbf{A}' && \text{(Assumption A4)} \\
 &= \sigma^2\mathbf{A}\mathbf{A}' \\
 &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
 &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \\
 \hat{V}(\hat{\beta}|\mathbf{X}) &= s_e^2(\mathbf{X}'\mathbf{X})^{-1} && \text{(Standard Error of } \hat{\beta})
 \end{aligned}$$

Standard Errors in R: Connecting to the Formula

```
mod <- lm(prestige ~ education + income, data=Prestige)

# The formula from the slides:  $s^2 (X'X)^{-1}$ 
sigma(mod)^2 * solve(t(X) %*% X)

# What R reports:
vcov(mod)

# Standard errors = sqrt of the diagonal
sqrt(diag(vcov(mod)))
coef(summary(mod))[, "Std. Error"] # identical
```

Every standard error in `summary(lm())` comes from $\sqrt{\text{diag}(s_e^2(\mathbf{X}'\mathbf{X})^{-1})}$.

Splitting y into Signal and Noise

- Every observation is part signal, part noise. Projection lets us split them:

$$y = \underbrace{Py}_{\text{signal } \hat{y}} + \underbrace{My}_{\text{noise } \hat{e}}$$

- The two pieces are **orthogonal**: $\hat{y}'\hat{e} = (Py)'(My) = y'PM y = 0$.
- So their squared lengths add up (Pythagorean theorem in $\mathbb{R}^n$):

$$\begin{aligned}\|y\|^2 &= \|\hat{y}\|^2 + \|\hat{e}\|^2 \\ \sum_{i=1}^n Y_i^2 &= \sum_{i=1}^n \hat{Y}_i^2 + \sum_{i=1}^n \hat{e}_i^2\end{aligned}$$

- How much of the total length is signal? That question leads to R^2 .

Analysis of Variance

- Subtract $\bar{Y}$ from both sides of the decomposition:

$$\mathbf{y} - \mathbf{1}\bar{Y} = (\hat{\mathbf{y}} - \mathbf{1}\bar{Y}) + \hat{\mathbf{e}}$$

- Take inner products. The cross terms vanish when $\mathbf{X}$ contains a constant:

$$(\hat{\mathbf{y}} - \mathbf{1}\bar{Y})' \hat{\mathbf{e}} = \hat{\mathbf{y}}' \hat{\mathbf{e}} - \bar{Y} \mathbf{1}' \hat{\mathbf{e}} = 0 - 0 = 0$$

- So we get the **analysis of variance** (ANOVA) decomposition:

$$\begin{aligned} \|\mathbf{y} - \mathbf{1}\bar{Y}\|^2 &= \|\hat{\mathbf{y}} - \mathbf{1}\bar{Y}\|^2 + \|\hat{\mathbf{e}}\|^2 \\ \underbrace{\sum (Y_i - \bar{Y})^2}_{\text{SST}} &= \underbrace{\sum (\hat{Y}_i - \bar{Y})^2}_{\text{SSR (regression)}} + \underbrace{\sum \hat{e}_i^2}_{\text{SSE (error)}} \end{aligned}$$

Goodness-of-Fit: R^2

- Dividing through by SST:

$$1 = \frac{SSR}{SST} + \frac{SSE}{SST}$$
$$R^2 \equiv \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- R^2 is the proportion of the variance in Y that is linearly explained by $\mathbf{X}$.
- $0 \leq R^2 \leq 1$: equals 0 when $\mathbf{X}$ explains nothing; equals 1 when $\hat{e}_i = 0$ for all i .
- We can always make $R^2 = 1$ by adding enough linearly independent columns to $\mathbf{X}$ (one per observation). This does not mean the model is good.

R^2 in Practice

- Typical values of R^2 vary by field:
 - Cross-sectional micro data (e.g. earnings regressions): $R^2 \approx 0.2$ – 0.4
 - Aggregate time-series (e.g. GDP growth models): $R^2 \approx 0.7$ – 0.9
 - Experimental data with noisy outcomes: R^2 can be very low and the treatment effect still precisely estimated.
- In R: `summary(lm(y ~ x))$r.squared` reports R^2 .
- The **adjusted** $R^2 = 1 - \frac{n-1}{n-k}(1 - R^2)$ penalizes adding regressors and can decrease when a variable adds no explanatory power.

Naming Conventions: A Warning

- Different textbooks use SSE and SSR with *opposite* meanings:

	Hansen / this course	Some other texts
$\frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum \hat{e}_i^2}$	SSR (regression)	ESS or SSR
	SSE (error)	RSS or SSE

- The underlying math is always the same: $SST = (\text{explained}) + (\text{unexplained})$.
- When reading papers or other textbooks, check which convention is in use.

The Cauchy–Schwarz Inequality

Cauchy–Schwarz Inequality

For any two random variables U and V with finite variances:

$$|\text{Cov}(U, V)| \leq \text{sd}(U) \text{sd}(V)$$

with equality if and only if $V = aU + b$ for some constants a, b .

- Dividing both sides by $\text{sd}(U) \text{sd}(V)$ gives $-1 \leq \text{corr}(U, V) \leq 1$, with equality only if the relationship is perfectly linear.
- This is an **algebraic fact**, not a statistical assumption. It holds for *any* joint distribution.

Cauchy–Schwarz Bounds the BLP Slope

- Since $\beta = \text{Cov}(X, Y)/\text{Var}(X)$, Cauchy–Schwarz gives us

$$|\beta| = \frac{|\text{Cov}(X, Y)|}{\text{Var}(X)} \leq \frac{\text{sd}(X) \text{sd}(Y)}{\text{Var}(X)} = \frac{\text{sd}(Y)}{\text{sd}(X)}.$$

- The BLP slope is bounded by the ratio of standard deviations.
- When $\text{Var}(Y) = \text{Var}(X)$, this simplifies to $|\beta| \leq 1$.

Cauchy–Schwarz Applied: When $\text{Var}(Y) = \text{Var}(X)$

- Suppose Y and X have the **same marginal variance**: $\text{Var}(Y) = \text{Var}(X)$.
- Then the BLP slope simplifies:

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\text{Cov}(X, Y)}{\text{sd}(X) \text{sd}(Y)} = \text{corr}(X, Y)$$

By Cauchy–Schwarz: $|\beta| \leq 1$, with equality only if $Y = aX + b$.

- If we also assume $\mathbb{E}[Y] = \mathbb{E}[X] = \mu$, then $\alpha = (1 - \beta)\mu$ and the BLP becomes:

Regression to the Mean

$$P(Y | X) = \mu + \beta(X - \mu), \quad |\beta| < 1$$

The predicted deviation from the mean is *always shrunk* by a factor β .

Regression to the Mean: The Setup

- Francis Galton (1886) measured fathers' and sons' heights. Tall fathers had sons who were tall—but *less* tall than their fathers. Short fathers had sons who were short—but *less* short.
- This is a natural result of the equal-variance projection we just derived: when $\text{Var}(Y) = \text{Var}(X)$, the slope is the correlation, and $|\beta| < 1$.
- The same structure appears in sample quantities. We proved $\text{SST} = \text{SSR} + \text{SSE}$. Divide by n :

$$\frac{1}{n} \sum (Y_i - \bar{Y})^2 = \frac{1}{n} \sum (\hat{Y}_i - \bar{Y})^2 + \frac{1}{n} \sum \hat{e}_i^2$$

$$\text{Var}(Y) = \text{Var}(\hat{Y}) + \text{Var}(\hat{e})$$

- So the variance of the fitted values is *smaller* than the variance of the raw data—some variance is “left behind” in the residuals.

Regression to the Mean: How Much Shrinkage?

- How much smaller? Recall $R^2 = SSR/SST$, so $SSR = R^2 \cdot SST$:

$$\frac{1}{n} \sum (\hat{Y}_i - \bar{Y})^2 = R^2 \cdot \frac{1}{n} \sum (Y_i - \bar{Y})^2 \quad (\text{divide both sides by } n)$$

$$\text{Var}(\hat{Y}) = R^2 \cdot \text{Var}(Y)$$

- When $R^2 < 1$, fitted values have **less spread** than the raw data—they are shrunk toward $\bar{Y}$. The noisier the relationship, the more shrinkage.
- Example: if $R^2 = 0.5$, the fitted values have half the variance of the data.

The Regression Fallacy

- We showed $\beta < 1$ under the assumption that $\text{Var}(Y) = \text{Var}(X)$ (a *stable* distribution).
- **Common error:** observing $\beta < 1$ and concluding that $\text{Var}(Y) < \text{Var}(X)$ —that the distribution is *converging*.
- But from the projection $Y = \alpha + \beta X + e$ with $\text{Cov}(X, e) = 0$:

$$\text{Var}(Y) = \beta^2 \text{Var}(X) + \text{Var}(e)$$

$\text{Var}(Y) < \text{Var}(X)$ requires $\beta^2 < 1 - \text{Var}(e)/\text{Var}(X)$, which is much stronger than $|\beta| < 1$.

- The variance of the projection error e **replenishes the spread** that regression to the mean appears to remove.

Examples of Regression to the Mean

- A student scores in the 95th percentile on Midterm 1, then the 80th on Midterm 2. Did they get lazy?
 - The “Sports Illustrated jinx” (athletes on the cover then underperform)
 - “Sophomore slumps” in politics, sports, music
 - Why the worst-performing schools improve and the best-performing schools decline, even without any intervention
- In 1933, Horace Secrist published *The Triumph of Mediocrity in Business*, documenting that department stores sorted by 1920 profits “converged toward mediocrity” over the next decade. There was no discovery—regression to the mean is a necessary feature of stable distributions.

Why This Matters for Research

- Regression to the mean is a threat to any research design that selects on the outcome or a noisy proxy of it:
 - Evaluating a tutoring program given to students who failed a test
 - Assessing a policy intervention targeted at the worst-performing hospitals
 - Studying whether “crackdowns” reduce crime in high-crime areas
- In each case, the group would have improved on average *even without the treatment*, because you selected them at an extreme.
- This is why we need control groups, and why “before–after” comparisons on selected samples are unreliable.